



8185 - A pocket of last resistance: characterizing the evaporating globule near a massive O-star binary

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				

Folder	Observation	Label	Observing Template	Science Target
	1	NIRSPEC	NIRSpec IFU Spectroscopy	(1) Pismis24_globule1_NIRSPEC
	5	NIRSPEC_bg	NIRSpec IFU Spectroscopy	(2) Pismis24_globule1_NIRSPEC_bg
	3	MIRI	MIRI Medium Resolution Spectroscopy	(5) Pismis24_globule1_MIRI
	4	MIRI_bg	MIRI Medium Resolution Spectroscopy	(8) Pismis24_globule1_MIRI_bkg

ABSTRACT

Stellar feedback, in the form of UV radiation from massive stars, gradually destroys the molecular cloud the stars formed in. The last pockets of last resistance against the progressing ionization front are the densest cores in the molecular cloud. These pre- and protostellar cores become exposed to the UV radiation when the lower density material around them evaporates and they become visible as evaporating globules. The mass of the forming proto stars in these globules as their protostellar envelopes are evaporated by the UV field. We propose NIRSPEC and MIRI/MRS spectroscopy of a large evaporating globule located very close to one of the most massive binaries in our Galaxy, Pis24_1. The exceptionally high levels of FUV and EUV radiation incident on this globule make it a unique case. These observations will provide a benchmark for studying the physical and chemical properties, as well as the evolution of the evaporating wind, under extreme EUV and FUV irradiation — far beyond the reach of current modeling efforts. Moreover, the proposed spectra will help us identify the nature of the various observed components within this globule, allowing us to derive mass-loss rates and the life time of the globule and the proto-planetary disks harbored inside.

OBSERVING DESCRIPTION

We propose NIRSPEC G235H and G395H integral field spectroscopy and MIRI/MRS integral field spectroscopy of an evaporating globule very close to one of the most massive binaries in our galaxy. The spatially resolved spectra will be used to determine the density and temperature of the neutral as well as the ionized emission from the evaporating flow.

The globule is expected to fill the entire field of view from NIRSPEC IFU and MIRI/MRS. We will obtain a dedicated background observations close by to correct for the telescope background as well as the emission from the surrounding HII region and the diffuse PDR emission.

Simultaneous MIRI imaging will be used to take Mid-infrared images of the Pismis 24 cluster population. Such simultaneous imaging enabled the discovery of this globule in the first place. The simultaneous imaging of the background observations will be used to image the globule at longer wavelength (F1130W, F1280W and F1500W).

For NIRSPEC we use the NRSRAPID readout and 5 groups to avoid saturation of the bright emission lines with a dither cycle pattern of 9 positions to improve the SNR in the fainter lines and improve the spatial sampling of the PSF. We use a LEAKCAL observation to correct for the emission

from the bright stars in Pismis 14 which fall in the MSA field of view. For MIRI/MRS we use the extended source dither pattern with 4 positions and 20 groups to reach the required SNR.

Proposal 8185 - Targets - A pocket of last resistance: characterizing the evaporating globule near a massive O-star binary

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Pismis24_globule1_NIRSPEC	RA: 17 24 43.8219 (261.1825912d) Dec: -34 11 53.33 (-34.19815d) Equinox: J2000		
	Comments: Category=Star Description=[Proplyds] Extended=YES				
	(2)	Pismis24_globule1_NIRSPEC_bg	RA: 17 24 42.3395 (261.1764146d) Dec: -34 11 54.23 (-34.19840d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES				
	(5)	Pismis24_globule1_MIRI	RA: 17 24 43.8380 (261.1826583d) Dec: -34 11 53.72 (-34.19826d) Equinox: J2000		
	Comments: Category=Star Description=[Proplyds] Extended=YES				
	(8)	Pismis24_globule1_MIRI_bkg	RA: 17 24 37.9797 (261.1582487d) Dec: -34 12 30.05 (-34.20835d) Equinox: J2000		
	Comments: Category=Star Description=[Proplyds] Extended=YES				

Proposal 8185 - Observation 1 - A pocket of last resistance: characterizing the evaporating globule near a massive O-star binary

Observation	Proposal 8185, Observation 1: NIRSPEC												Fri Jan 16 19:01:07 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	Pismis24_globule1_NIRSPEC	RA: 17 24 43.8219 (261.1825912d) Dec: -34 11 53.33 (-34.19815d) Equinox: J2000										
	Comments: Category=Star Description=[Proplyds] Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		SMALL		1		9					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8		
	2	G395H/F290LP	NRSIRS2RAPID	5	1	true	true	NONE	9	9	787.8		
	3	G235H/F170LP	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8		
	4	G235H/F170LP	NRSIRS2RAPID	5	1	true	true	NONE	9	9	787.8		
Special Requirements	Sequence Observations 1, 5, Non-interruptible												

Proposal 8185 - Observation 5 - A pocket of last resistance: characterizing the evaporating globule near a massive O-star binary

Observation	Proposal 8185, Observation 5: NIRSPEC_bg												Fri Jan 16 19:01:07 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	Pismis24_globule1_NIRSPEC_bg	RA: 17 24 42.3395 (261.1764146d) Dec: -34 11 54.23 (-34.19840d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	CYCLING		LARGE		1		9					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8		
	2	G395H/F290LP	NRSIRS2RAPID	5	1	true	true	NONE	9	9	787.8		
	3	G235H/F170LP	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8		
	4	G235H/F170LP	NRSIRS2RAPID	5	1	true	true	NONE	9	9	787.8		
Special Requirements	Sequence Observations 1, 5, Non-interruptible												

Proposal 8185 - Observation 3 - A pocket of last resistance: characterizing the evaporating globule near a massive O-star binary

Observation	Proposal 8185, Observation 3: MIRI													Fri Jan 16 19:01:07 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI_bg (Obs 4)]													
Diagnostics	(MIRI (Obs 3)) Warning (Form): Filter mismatch between science and background observations may result in incorrect background subtraction.													
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(Visit 3:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(5)	Pismis24_globule1_MIRI	RA: 17 24 43.8380 (261.1826583d) Dec: -34 11 53.72 (-34.19826d) Equinox: J2000											
	Comments: Category=Star Description=[Proplyds] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
	F560W	Channel 1				YES			FULL		Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F560W	FASTR1	20	2	1	Dither 1	4	8	455.107		
	1	SHORT(A)	MRSLONG		FASTR1	20	2	1	Dither 1	4	8	455.107		
	1	SHORT(A)	MRSSHORT		FASTR1	20	2	1	Dither 1	4	8	455.107		
	2		IMAGER	F770W	FASTR1	20	2	1	Dither 1	4	8	455.107		
	2	MEDIUM(B)	MRSLONG		FASTR1	20	2	1	Dither 1	4	8	455.107		
	2	MEDIUM(B)	MRSSHORT		FASTR1	20	2	1	Dither 1	4	8	455.107		
	3		IMAGER	F1000W	FASTR1	20	2	1	Dither 1	4	8	455.107		
	3	LONG(C)	MRSLONG		FASTR1	20	2	1	Dither 1	4	8	455.107		
	3	LONG(C)	MRSSHORT		FASTR1	20	2	1	Dither 1	4	8	455.107		

Proposal 8185 - Observation 3 - A pocket of last resistance: characterizing the evaporating globule near a massive O-star binary

Special Requirements	Aperture PA Range 260.0 to 300.0 Degrees (V3 260.0 to 300.0) Sequence Observations 4, 3 (reordered), Non-interruptible
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Proposal 8185 - Observation 4 - A pocket of last resistance: characterizing the evaporating globule near a massive O-star binary

Observation	Proposal 8185, Observation 4: MIRI_bg													Fri Jan 16 19:01:07 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [MIRI (Obs 3)]													
Diagnostics	(MIRI_bg (Obs 4)) Warning (Form): Imager Filter overlap.													
	(MIRI_bg (Obs 4)) Warning (Form): Imager Filter overlap.													
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(Visit 4:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous				
	(8)	Pismis24_globule1_MIRI_bkg	RA: 17 24 37.9797 (261.1582487d) Dec: -34 12 30.05 (-34.20835d) Equinox: J2000											
	Comments: Category=Star Description=[Proplyds] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
		All MRS				YES			FULL		Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				BACKGROUND				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F1130W	FASTR1	20	2	1	Dither 1	4	8	455.107		
	1	SHORT(A)	MRSLONG		FASTR1	20	2	1	Dither 1	4	8	455.107		
	1	SHORT(A)	MRSSHORT		FASTR1	20	2	1	Dither 1	4	8	455.107		
	2		IMAGER	F1280W	FASTR1	20	2	1	Dither 1	4	8	455.107		
	2	MEDIUM(B)	MRSLONG		FASTR1	20	2	1	Dither 1	4	8	455.107		
	2	MEDIUM(B)	MRSSHORT		FASTR1	20	2	1	Dither 1	4	8	455.107		
	3		IMAGER	F1500W	FASTR1	20	2	1	Dither 1	4	8	455.107		
	3	LONG(C)	MRSLONG		FASTR1	20	2	1	Dither 1	4	8	455.107		
	3	LONG(C)	MRSSHORT		FASTR1	20	2	1	Dither 1	4	8	455.107		

Proposal 8185 - Observation 4 - A pocket of last resistance: characterizing the evaporating globule near a massive O-star binary	
Special Requirements	Aperture PA Range 260.0 to 300.0 Degrees (V3 260.0 to 300.0) Sequence Observations 4, 3 (reordered), Non-interruptible